

SUN ZHELUN

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🎓 EDUCATION

Tsinghua University, Beijing, China 2023 – Present

Ph.D. Candidate in Ecology

Research focus: Intelligent systems and scientific workflows for remote sensing analysis

China Agricultural University, Beijing, China 2019 – 2023

B.Sc. in Geographic Information Science

GPA: **3.85 / 4.00**

Relevant coursework: Programming, Data Structures, Spatial Analysis and Geostatistics, Remote Sensing, Python GIS Development

📄 PUBLICATIONS

First-author publications

- **Sun, Z.**, Zhou, Y., & Yang, J. (2026). An LLM-based multi-agent system for remote sensing analysis. *Big Earth Data*. doi:10.1080/20964471.2025.2600178
- **Sun, Z.**, Li, X., Wei, H., Feng, Z., & Yang, J. (2024). Landsat image classification using a deep learning model and multiple-source training samples. *International Journal of Digital Earth*, 17(1), 2409351. doi:10.1080/17538947.2024.2409351
- **Sun, Z.**, Xiao, B., Zhao, Z., Yu, G., Cao, Q., Wang, M., & Yao, X. (2023). Loading and service implementation of GF-1 satellite data based on Open Data Cube. *Remote Sensing Technology and Application*, 38(4), 794–802. doi:10.11873/j.issn.1004-0323.2023.4.0794

Co-author publications

- Zhang, Y., **Sun, Z.**, Ji, J. S., Li, X., & Yang, J. (2026). Measuring perceived green volume for quantifying urban green exposure. *Urban Forestry & Urban Greening*, 116, 129231. doi:10.1016/j.ufug.2025.129231

🔧 SELECTED RESEARCH EXPERIENCE

LLM-based Remote Sensing Agent System 2024 – Present

- Developed a user-centric framework for remote sensing analysis, leveraging LLMs as the reasoning core and natural-language interface for non-expert users.
- Built ExpertsRS, a prototype multi-agent system based on AutoGen, with distinct agents coordinating remote sensing data, tools, and reasoning processes.
- Demonstrated the potential of the system to lower technical barriers in remote sensing analysis through natural-language interaction and interpretable result generation.

Structured Research Asset Prototype for Urban Forest Remote Sensing 2025 – Present

- Developed a structured intermediate layer to connect urban forest monitoring objectives with remote sensing variables, indicators, and data sources.
- Organized domain knowledge into an agent-oriented representation to support information retrieval and workflow assistance in urban forest biodiversity mapping.
- Built an early-stage prototype for linking remote sensing variables, ecological indicators, and stage-wise analytical processes as reusable research assets.

GF-1 Data Loading and Service Development Based on Open Data Cube 2020 – 2022

- Developed Python scripts for loading GF-1 satellite data into the Open Data Cube (ODC) framework.
- Implemented a web-based data management and service workflow for accessing and analyzing loaded GF-1 imagery.
- Built a prototype water detection service using the ODC Web UI.

Nighttime Light Analysis of the Ukraine–Russia War

2022

- Conducted a remote sensing study using VIIRS nighttime light data to assess war-related impacts on human activity.
- Produced pre-war and wartime nighttime light maps of Ukraine and quantified light loss as a proxy for local disruption.

🏆 HONORS AND GRANTS

Taihu Zhiguang Scholarship, Tsinghua University	2025
Excellent Graduate, Beijing Municipality	2023
First Prize Scholarship, China Agricultural University	2021
PI, National Undergraduate Training Program for Innovation (RMB 15,000)	2020

⚙️ SKILLS

- **AI and Scientific Computing:** Python, PyTorch, scikit-learn, pandas, NumPy, SciPy, matplotlib
- **Remote Sensing and Geospatial Computing:** Google Earth Engine, GDAL, rasterio, ArcGIS, ENVI
- **Programming and Data Infrastructure:** SQL, C#, JavaScript, NetworkX
- **Research and Development Tools:** LaTeX, Linux (Ubuntu), Git, Conda